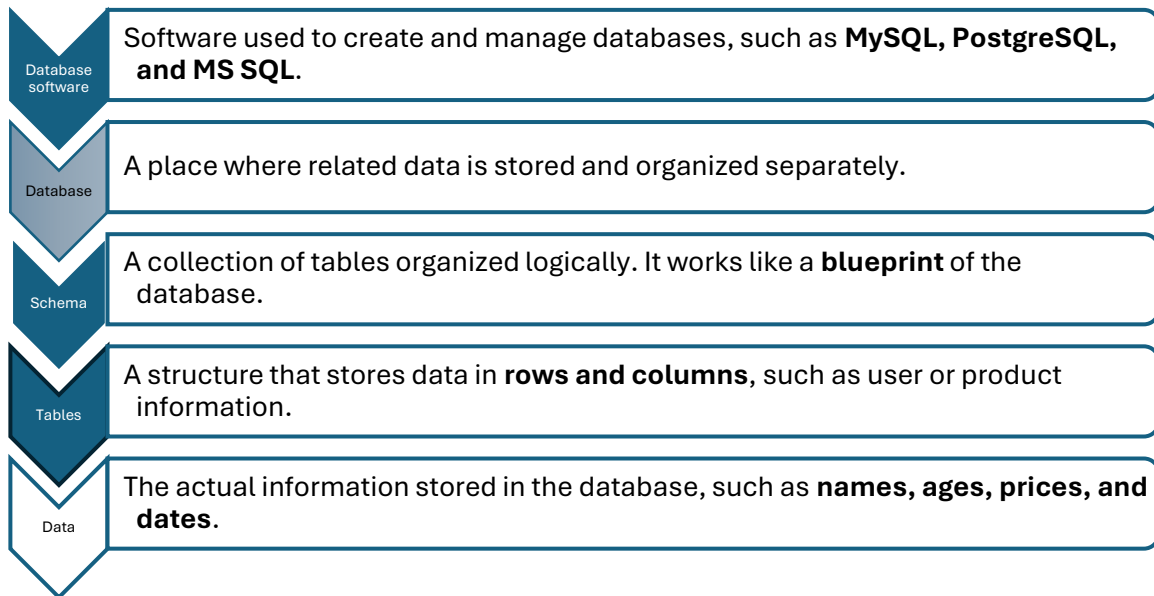


Introduction To MYSQL

SQL is built by 'Database'. To communicate the Database by "MySQL". Oracle, etc., "MySQL" is the base language. MySQL is stored in Structured of "Rows" and "Columns" like a Table.

General Database Structure



There are different types of MySQL:

DDL(Data definition language)

Purpose: Defines, modifies, and alters the actual structure or schema of the database.

Core Commands:

- CREATE (makes new tables)
- ALTER (modifies table structures)
- DROP (deletes an entire table permanently)
- TRUNCATE (removes all records but keeps the structure)

DML(Data Manipulation language)

Purpose: Manages and modifies the actual data rows stored inside the existing tables.

Core Commands:

- INSERT (adds new rows)
- UPDATE (edits existing data)
- DELETE (removes specific records).

TCL(Transaction Control Language)

Purpose: Manages transactions to group multiple tasks together, ensuring database integrity.

Core Commands:

- COMMIT (saves changes permanently)
- ROLLBACK (undoes changes if an error occurs)

Datatypes of SQL:

NUMERIC	DATE AND TIME	STRING
int	date and time("Y:M:D")	varchar
bigint	timestamp (change in universal time)	char
float		

Movie Database Management and Analytical Reporting Case Study:

#Create a database and use the database

```
create database Movies_DB;

use Movies_DB;

create table movie
(
rel_year int,
mov_name varchar(100),
act_name varchar(50),
lang varchar(30),
dir_name varchar(100)
);
```

#Insert the data's

```
INSERT INTO MOVIES VALUES (1960,'Kalathur Kannamma','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES VALUES (1962,'Kannum Karalum','Kamal Haasan','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES VALUES (1962,'Paadha Kaanikkai','Kamal Haasan','Tamil','K. Shankar');
INSERT INTO MOVIES VALUES (1962,'Paarthaal Pasi Theerum','Kamal Haasan','Tamil','A. Bhimsingh');
INSERT INTO MOVIES VALUES (1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy');
INSERT INTO MOVIES VALUES (1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan');
INSERT INTO MOVIES VALUES (1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam');
INSERT INTO MOVIES VALUES (1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan');
INSERT INTO MOVIES VALUES (1971,'Anubhavangal Paalichakal','Mammootty','Malayalam','K. S. Sethumadhavan');
INSERT INTO MOVIES VALUES (1972,'Kurathi Magan','Kamal Haasan','Tamil','K. S. Gopalakrishnan');
INSERT INTO MOVIES VALUES (1973,'Arangetram','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES VALUES (1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan');
INSERT INTO MOVIES VALUES (1973,'Sollathan Ninaikkiren','Kamal Haasan','Tamil','K. Balachander');
INSERT INTO MOVIES VALUES (1974,'Anbu Thangai','Kamal Haasan','Tamil','S. P. Muthuraman');
INSERT INTO MOVIES VALUES (1974,'Aval Oru Thodar Kathai','Kamal Haasan','Tamil','K. Balachander');
```

Retrieve all details from the movie table.

```
select * from MOVIES;
```

Display distinct movie languages.

```
select distinct lang from movies;
```

Retrieve movies acted by a specific actor "Jayam Ravi".

```
select mov_name from movies where act_name = "Jayam Ravi";
```

Sort movies by release year in ascending order.

```
select mov_name from movies order by rel_year;
```

Retrieve the latest 10 movies based on release year.

```
select mov_name,rel_year from movies order by rel_year desc limit 10;
```

Count the total number of movies.

```
SELECT COUNT(mov_name) AS total_movies FROM movies;
```

List languages with more than 100 movies.

```
SELECT lang, COUNT(mov_name) AS movie_count FROM movies GROUP BY lang HAVING COUNT(mov_name) > 100;
```

Identify actors who have worked in more than three languages.

```
SELECT act_name, COUNT(DISTINCT lang) AS unique_languages FROM movies GROUP BY act_name HAVING COUNT(DISTINCT lang) > 3;
```